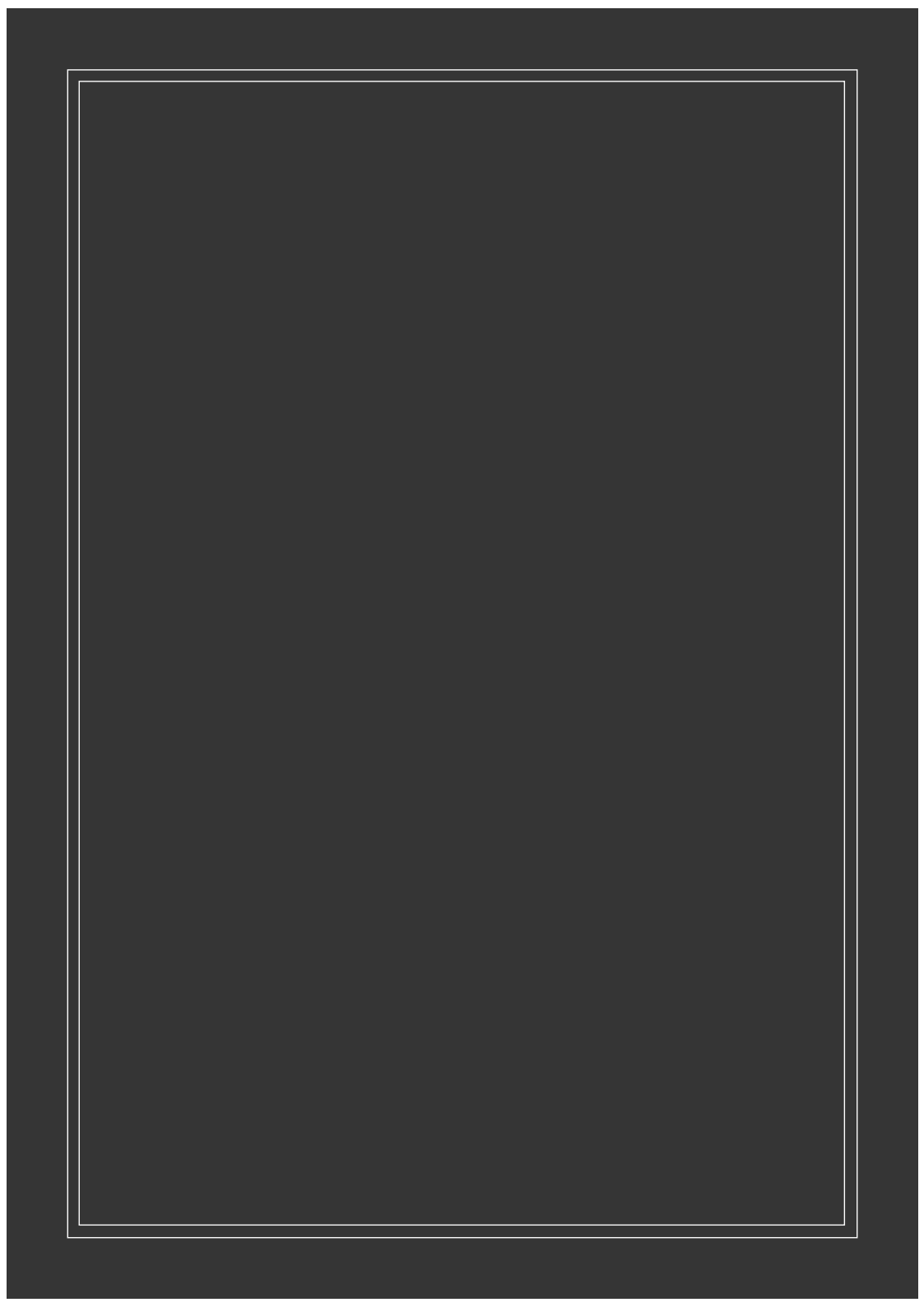


AN ELEMENTARY INTRODUCTION TO
MICROECONOMICS



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KUTUSH KUMAR, KUTUM PAKHI

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TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches An Elementary Introduction to Microeconomics one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

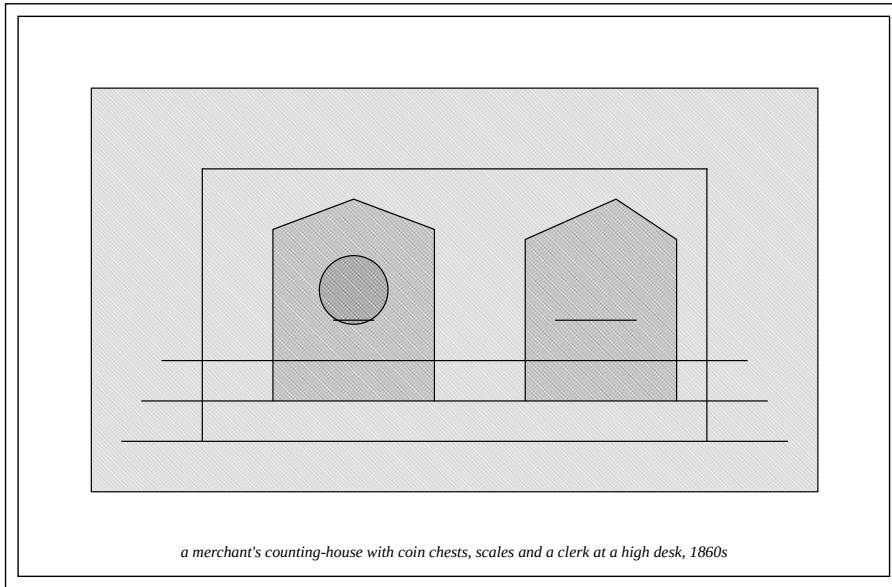
A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.



CHAPTER I

What Finance Is

A practical question about what finance is



This chapter introduces what finance is for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

reasoning should use the facts given in the question and keep timing and units consistent.

13.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

13.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

13.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

13.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XIV · WORKING CAPITAL AND SHORT-TERM FINANCING

14.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

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14.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

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CHAPTER XV · LONG-TERM FINANCING AND THE CAPITAL MARKET

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A calculation can be correct and still answer the wrong question. For finance in plain words, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

1.2 FINANCE BESIDE ITS NEIGHBOURING SUBJECTS

The purpose here is to make finance beside its neighbouring subjects concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating efficiency** means How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sole proprietorship** means A business owned and run by one person, who keeps all profit and answers for all debts. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Partnership** means A business owned by two or more people who agree to run it together and share its profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For finance beside its neighbouring subjects, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

The business meaning matters as much as the arithmetic.

This section builds the idea of finance beside its neighbouring subjects before moving to application.

Operating efficiency. How much useful output a business gets from the money and effort it uses.

The term operating efficiency gives the chapter a precise name for this part of the problem.

Sole proprietorship. A business owned and run by one person, who keeps all profit and answers for all debts.

The term sole proprietorship gives the chapter a precise name for this part of the problem.

Partnership. A business owned by two or more people who agree to run it together and share its profit.

The term partnership gives the chapter a precise name for this part of the problem.

The central question in finance beside its neighbouring subjects is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

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CHAPTER IX · RISK AND RETURN

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CHAPTER X · LEVERAGE

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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CHAPTER IV · INTEREST AND FUTURE VALUE

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CHAPTER V · PRESENT VALUE AND DISCOUNTING

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CHAPTER VI · ANNUITIES

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SUMMARY

The central purpose of What Finance Is is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

The chapter has connected its main terms, method, and business decision.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

Finance becomes stronger when a number is followed by a question about meaning.

WHERE PEOPLE GO WRONG

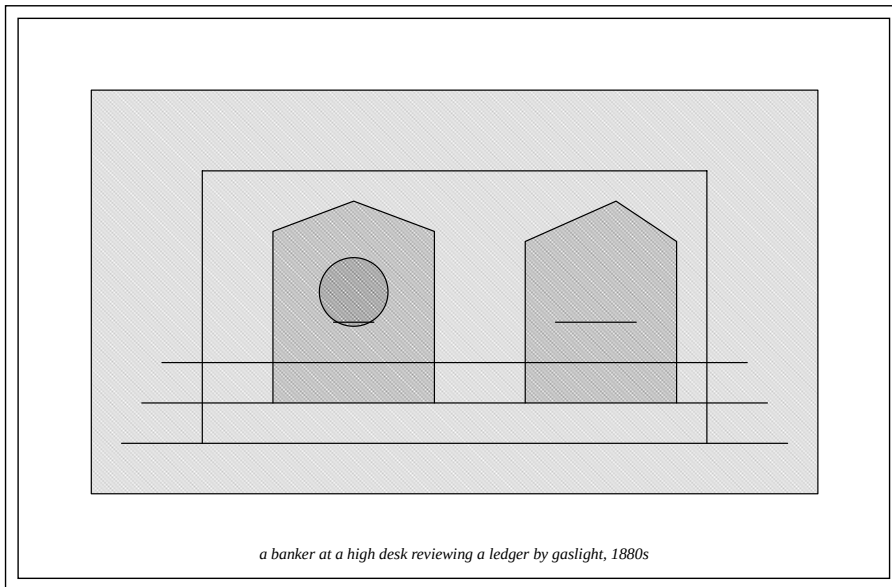
Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

Most errors arise from mixing terms, timing, signs, or decision rules.

IN HISTORY

The Dutch East India Company (VOC), chartered 1602, sold shares to the public that traded in Amsterdam, an early company separating owners from managers. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

A historical case shows how the chapter's financial problem appeared in practice.



CHAPTER II

The Financial Manager's Decisions

A practical question about the financial manager's decisions



This chapter introduces the financial manager's decisions for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Risk premium. The extra return demanded for taking risk instead of a risk-free investment. (*Chapter XI*)

Risk-free rate. The return on an investment with no risk of loss, such as a short-term government security. (*Chapter XI*)

Sale-and-leaseback. A deal in which a firm sells an asset it owns and leases it back from the buyer. (*Chapter XVI*)

Security market line. The graph of the capital asset pricing model, with beta on one axis and required return on the other. (*Chapter XI*)

Semiannual interest. Interest paid twice a year, each payment being half the yearly amount. (*Chapter XII*)

Share price. The price at which one share of a company can be bought or sold today. (*Chapter III*)

Shareholder. A person who owns part of a company by holding its shares. (*Chapter III*)

Short-term financing. Money borrowed to be repaid within one year. (*Chapter XIV*)

Simple interest. Interest calculated only on the principal sum, so it is the same every period. (*Chapter IV*)

Sole proprietorship. A business owned and run by one person, who keeps all profit and answers for all debts. (*Chapter I*)

Stakeholder. Anyone affected by a firm's actions, such as staff, customers, lenders and the community. (*Chapter III*)

Standard deviation. The square root of the variance; it shows how far results typically stray from the expected value. (*Chapter IX*)

Sunk cost. Money already spent that cannot be recovered whatever you decide now. (*Chapter VII*)

Systematic risk. Risk that affects all investments, such as a recession; it cannot be diversified away. (*Chapter XI*)

Tax shield. The tax saved because a cost, such as interest or depreciation, reduces taxable income. (*Chapter XVI*)

Term loan. A loan repaid in regular instalments over a fixed number of years, usually from a bank. (*Chapter XV*)

Terminal cash flow. The cash received or paid when a project ends, such as the sale of scrap. (*Chapter VII*)

Time value of money. The idea that a taka received today is worth more than a taka received later. (*Chapter IV*)

Timeline. A line marking each period, with the cash flow written at the point where it occurs. (*Chapter IV*)

Total leverage. The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. (*Chapter X*)

Trade credit. Credit from a supplier who lets a buyer pay some days after delivery. (*Chapter XIV*)

Treasurer. The officer in charge of handling cash, raising money and keeping relations with banks. (*Chapter II*)

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the three big decisions, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

2.2 WHAT A FINANCIAL MANAGER DOES

The purpose here is to make what a financial manager does concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend decision** means A choice about how much profit to pay out to owners and how much to keep for the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Treasurer** means The officer in charge of handling cash, raising money and keeping relations with banks. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Controller** means The officer in charge of accounting records, reports and checking that money is used as planned. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial planning** means Estimating the money a business will need and receive in future, and arranging how to meet the gap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of what a financial manager does before moving to application.

Dividend decision. A choice about how much profit to pay out to owners and how much to keep for the business..

The term dividend decision gives the chapter a precise name for this part of the problem.

Treasurer. The officer in charge of handling cash, raising money and keeping relations with banks..

The term treasurer gives the chapter a precise name for this part of the problem.

Controller. The officer in charge of accounting records, reports and checking that money is used as planned..

The term controller gives the chapter a precise name for this part of the problem.

Financial planning. Estimating the money a business will need and receive in future, and arranging how to meet the gap..

The term financial planning gives the chapter a precise name for this part of the problem.

The central question in what a financial manager does is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

Interest. The payment a borrower makes to a lender for the use of money, or the reward a saver earns. (Chapter IV)

Interest rate. The interest for one period written as a percentage of the sum, for example 8% a year. (Chapter IV)

Intermediate-term financing. Money borrowed to be repaid in more than one year but not more than about five. (Chapter XIV)

Internal rate of return. The discount rate at which a project's net present value is exactly zero. (Chapter VIII)

Inventory. The goods a business holds for sale, such as the books on a shop's shelves. (Chapter XIV)

Investment decision. A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. (Chapter II)

Lease. A contract in which the owner of an asset lets another party use it for regular payments. (Chapter XVI)

Lessee. The user who pays to use the asset under a lease. (Chapter XVI)

Lessor. The owner who lets the asset out under a lease. (Chapter XVI)

Leverage. The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. (Chapter X)

Liability. Money a business owes to others, such as a bank loan or an unpaid supplier bill. (Chapter II)

Long-term financing. Money raised for more than about five years, or with no fixed repayment date. (Chapter XIV)

Market return. The average return of all investments in a market during a period. (Chapter XI)

Matching principle. Pay for short-lived needs with short-term money and for lasting needs with long-term money. (Chapter XIV)

Maturity. The date on which a bond or loan must be repaid. (Chapter XII)

Mixed stream. A series of cash flows of different sizes in different periods. (Chapter V)

Mutually exclusive projects. Projects of which only one can be chosen, because choosing one rules out the other. (Chapter VII)

Net present value. The present value of all a project's future cash flows minus its initial investment. (Chapter VIII)

Net working capital. Current assets minus current liabilities; what remains for trading after short debts are paid. (Chapter XIV)

Operating cash flow. The cash a project brings in each year after paying its cash costs and tax. (Chapter VII)

Operating cycle. The days from buying stock to collecting cash from its sale. (Chapter XIV)

Operating efficiency. How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. (Chapter I)

Operating lease. A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. (Chapter XVI)

Operating leverage. The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the

later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the financial manager's decisions. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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Cash inflow. Money coming into a business or to an investor, shown as a positive number. (Chapter V)

Cash outflow. Money going out of a business or from an investor, shown as a negative number. (Chapter V)

Coefficient of variation. Standard deviation divided by expected value, giving risk per unit of expected return. (Chapter IX)

Collateral. An asset a lender may take if the borrower fails to repay. (Chapter XV)

Common stock. Shares that give owners a vote and a claim on profit after all other claims are met. (Chapter XIII)

Company. A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. (Chapter I)

Compound interest. Interest calculated on the principal sum and also on interest already earned. (Chapter IV)

Compounding period. The length of time after which interest is added to the balance, such as a year or six months. (Chapter IV)

Constant growth model. The dividend valuation model for dividends that grow at the same rate every year. (Chapter XIII)

Contribution margin. Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. (Chapter X)

Controller. The officer in charge of accounting records, reports and checking that money is used as planned. (Chapter II)

Cost of capital. The return a firm must earn on its investments to satisfy those who supply its money. (Chapter XV)

Coupon rate. The yearly interest of a bond, as a percentage of its face value. (Chapter XII)

Covenant. A condition in a loan agreement that the borrower promises to keep. (Chapter XV)

Debenture. A bond backed only by the issuer's general promise to pay, not by a particular asset. (Chapter XV)

Depreciation. The share of an asset's cost charged against income each year as the asset wears out. (Chapter VII)

Discount bond. A bond that sells for less than its face value. (Chapter XII)

Discount rate. The yearly percentage used in discounting; it states the return you could earn elsewhere. (Chapter V)

Discounted payback period. The payback time when each cash flow is first discounted to present value. (Chapter VIII)

Discounting. Finding present value by removing the interest a sum could earn before it arrives. (Chapter V)

Diversification. Spreading money across many investments so that one failure does less damage. (Chapter XI)

Dividend. A share of a company's profit paid out to its owners instead of being kept in the business. (Chapter II)

Dividend decision. A choice about how much profit to pay out to owners and how much to keep for the business. (Chapter II)

SUMMARY

The central purpose of The Financial Manager's Decisions is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

The chapter has connected its main terms, method, and business decision.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

Finance becomes stronger when a number is followed by a question about meaning.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

Most errors arise from mixing terms, timing, signs, or decision rules.

IN HISTORY

General Motors in the 1920s: Donaldson Brown, who had come from DuPont, built financial controls linking return on investment to decisions. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

A historical case shows how the chapter's financial problem appeared in practice.

REVIEW QUESTIONS

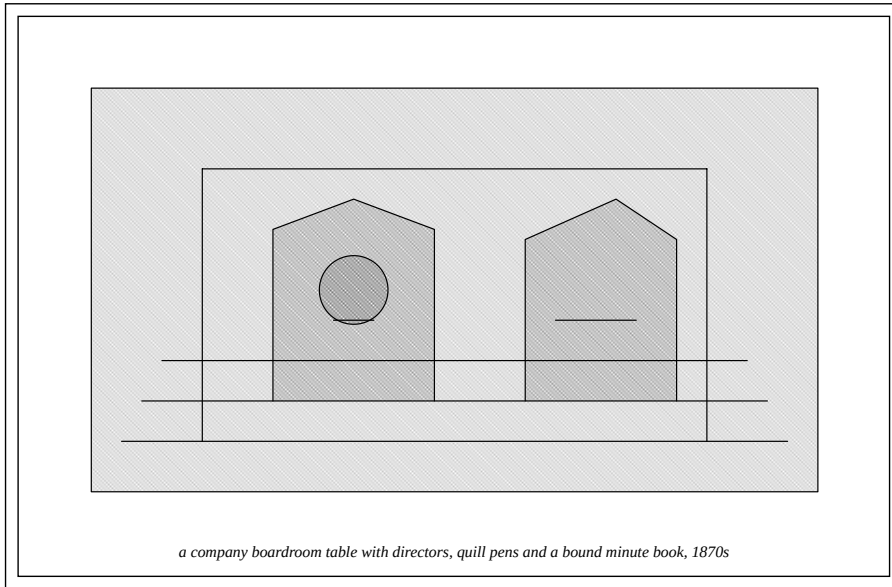
02.T1 name and tell apart the three decisions using a new business

02.T2 state the work of a treasurer and a controller

02.T3 build a balance sheet from a new list of items and check that it balances

PAUSE AND CHECK 02.P1 After the three decisions, state the rule in your own words.

PAUSE AND CHECK 02.P2 After the balance sheet, state the rule in your own words.



CHAPTER III

The Goal of the Firm and Agency Theory

A practical question about the goal of the firm and agency theory



This chapter introduces the goal of the firm and agency theory for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section gives the arithmetic rule used in the book.

The tiny example shows the order of work.

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A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A.1 A1.1 PERCENTAGES AND PERCENTAGE CHANGE

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A.2 A1.2 RATIOS AND AVERAGES

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A.3 A1.3 REARRANGING AN EQUATION AND ROUNDING

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

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A.4 A1.4 READING A GRAPH

Use the rule as a reference. A percentage expresses a part per hundred; a ratio compares two quantities; an average summarises a set of values; an equation can be rearranged by performing the same operation on both sides; rounding should wait until the final useful step. For a graph, read the axes, units, scale, and direction before interpreting the line or bar.

Given: the relevant numbers. Find: the requested quantity. Formula: write the relationship first. Steps: substitute the values and calculate. Answer: state the result with its unit. Check: compare the result with the size of the inputs.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For profit maximisation and its limits, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

3.2 WEALTH MAXIMISATION

The purpose here is to make wealth maximisation concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Stakeholder** means Anyone affected by a firm's actions, such as staff, customers, lenders and the community. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Principal** means A person who hires another to act for them. A shop owner who hires a manager is a principal. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Agent** means A person who acts for a principal and makes decisions for the principal. A hired shop manager is an agent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Agency problem** means A conflict that arises when an agent's own interests differ from those of the principal. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of wealth maximisation before moving to application.

Stakeholder. Anyone affected by a firm's actions, such as staff, customers, lenders and the community..

The term stakeholder gives the chapter a precise name for this part of the problem.

Principal. A person who hires another to act for them.

The term principal gives the chapter a precise name for this part of the problem.

Agent. A person who acts for a principal and makes decisions for the principal.

The term agent gives the chapter a precise name for this part of the problem.

Agency problem. A conflict that arises when an agent's own interests differ from those of the principal..

The term agency problem gives the chapter a precise name for this part of the problem.

The central question in wealth maximisation is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

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SUMMARY

The central purpose of Leasing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify

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SUMMARY

The central purpose of The Goal of the Firm and Agency Theory is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Berle and Means, The Modern Corporation and Private Property (1932), and Jensen and Meckling (1976) on agency costs. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

Tax shield. The tax saved because a cost, such as interest or depreciation, reduces taxable income..

The term tax shield gives the chapter a precise name for this part of the problem.

After-tax cash outflow. A payment's cost after subtracting the tax it saves..

The term after-tax cash outflow gives the chapter a precise name for this part of the problem.

The central question in the lease-or-buy decision is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Tax shield** means The tax saved because a cost, such as interest or depreciation, reduces taxable income. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **After-tax cash outflow** means A payment's cost after subtracting the tax it saves. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the lease-or-buy decision, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

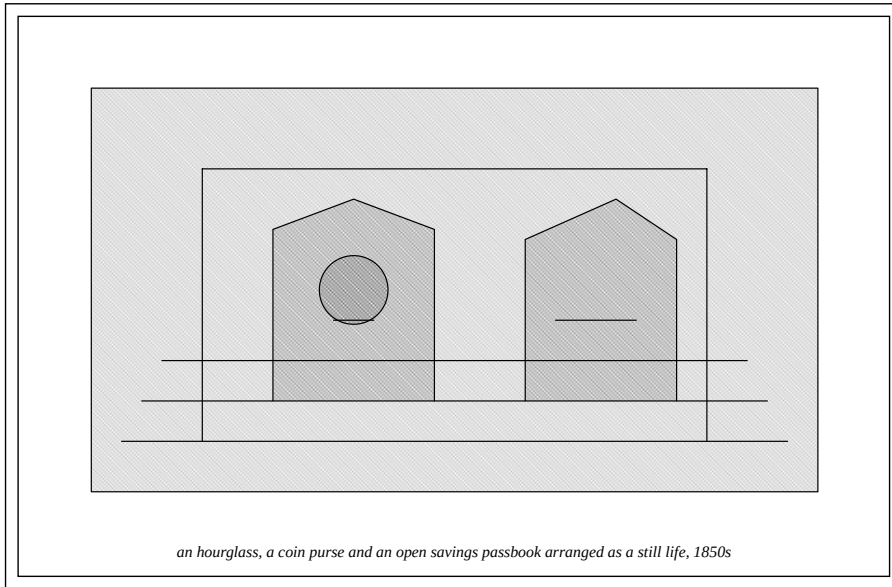
* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to leasing. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.



CHAPTER IV

Interest and Future Value

A practical question about interest and future value



This chapter introduces interest and future value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of what leasing is before moving to application.

Lease. A contract in which the owner of an asset lets another party use it for regular payments..

The term lease gives the chapter a precise name for this part of the problem.

Lessor. The owner who lets the asset out under a lease..

The term lessor gives the chapter a precise name for this part of the problem.

Lessee. The user who pays to use the asset under a lease..

The term lessee gives the chapter a precise name for this part of the problem.

The central question in what leasing is is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

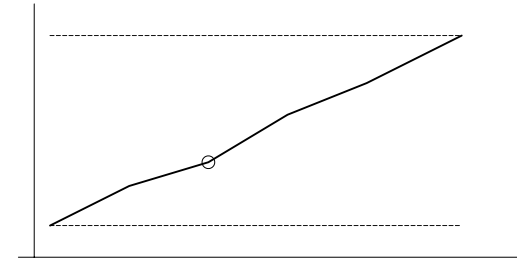


FIG. 16.1. Leasing figure 1

16.1 WHAT LEASING IS

The purpose here is to make what leasing is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Lease** means A contract in which the owner of an asset lets another party use it for regular payments. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Lessor** means The owner who lets the asset out under a lease. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Lessee** means The user who pays to use the asset under a lease. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what leasing is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a

In this chapter, **Interest** means The payment a borrower makes to a lender for the use of money, or the reward a saver earns. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Interest rate** means The interest for one period written as a percentage of the sum, for example 8% a year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why time changes the value of money, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

4.2 SIMPLE AND COMPOUND INTEREST

The purpose here is to make simple and compound interest concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Principal sum** means The original amount lent, borrowed or deposited, before any interest is added. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Simple interest** means Interest calculated only on the principal sum, so it is the same every period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Interest. The payment a borrower makes to a lender for the use of money, or the reward a saver earns..

The term interest gives the chapter a precise name for this part of the problem.

Interest rate. The interest for one period written as a percentage of the sum, for example 8% a year..

The term interest rate gives the chapter a precise name for this part of the problem.

The central question in why time changes the value of money is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of simple and compound interest before moving to application.

Principal sum. The original amount lent, borrowed or deposited, before any interest is added..

The term principal sum gives the chapter a precise name for this part of the problem.

Simple interest. Interest calculated only on the principal sum, so it is the same every period..

The term simple interest gives the chapter a precise name for this part of the problem.

before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Dhaka Stock Exchange: its share-price boom of 2010 and the sharp fall of early 2011. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

15.T1 sort given facts into short, intermediate and long-term financing

15.T2 describe the main sources of long-term money for a new company with their strengths and limits

15.T3 explain what retained earnings cost the firm

15.T4 compute the weighted average cost of capital for new figures

PAUSE AND CHECK 15.P1 After the features of financing, state the rule in your own words.

PAUSE AND CHECK 15.P2 After capital-market issues, state the rule in your own words.

A historical case shows how the chapter's financial problem appeared in practice.

In this chapter, **Factor table** means A table of ready-made multipliers that saves repeating a long calculation. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For future value of a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to interest and future value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when

Factor table. A table of ready-made multipliers that saves repeating a long calculation..

The term factor table gives the chapter a precise name for this part of the problem.

The central question in future value of a single sum is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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SUMMARY

The central purpose of Interest and Future Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

15.3 RAISING FUNDS IN THE CAPITAL MARKET

The purpose here is to make raising funds in the capital market concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Initial public offering** means A company's first sale of shares to the public. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Rights issue** means An offer of new shares to existing shareholders, in proportion to their holdings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Underwriter** means An institution that promises to buy any shares of a new issue that the public does not. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For raising funds in the capital market, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

15.4 THE COST OF CAPITAL

The purpose here is to make the cost of capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

This section builds the idea of raising funds in the capital market before moving to application.

Initial public offering. A company's first sale of shares to the public..

The term initial public offering gives the chapter a precise name for this part of the problem.

Rights issue. An offer of new shares to existing shareholders, in proportion to their holdings..

The term rights issue gives the chapter a precise name for this part of the problem.

Underwriter. An institution that promises to buy any shares of a new issue that the public does not..

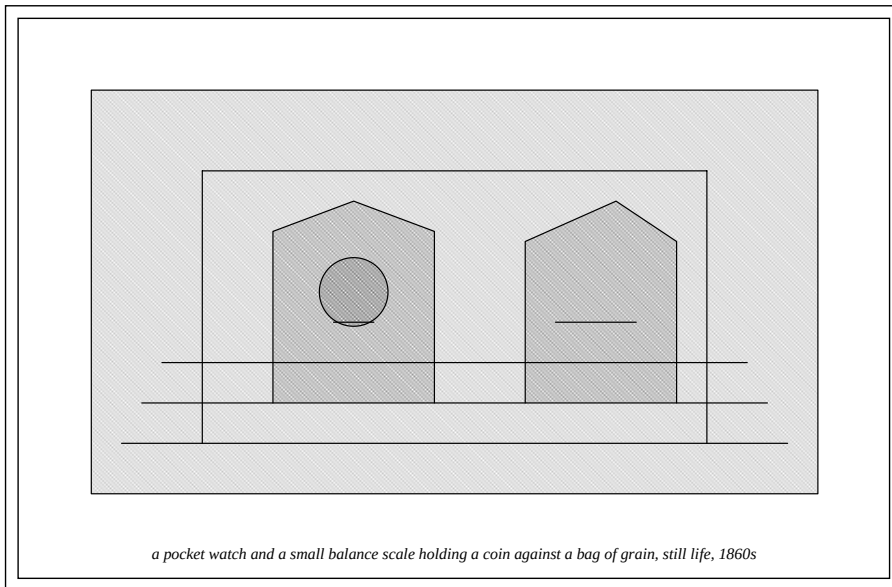
The term underwriter gives the chapter a precise name for this part of the problem.

The central question in raising funds in the capital market is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the cost of capital before moving to application.



CHAPTER V

Present Value and Discounting

A practical question about present value and discounting



This chapter introduces present value and discounting for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of term loans and the features of financing before moving to application.

Term loan. A loan repaid in regular instalments over a fixed number of years, usually from a bank..

The term term loan gives the chapter a precise name for this part of the problem.

Collateral. An asset a lender may take if the borrower fails to repay..

The term collateral gives the chapter a precise name for this part of the problem.

Covenant. A condition in a loan agreement that the borrower promises to keep..

The term covenant gives the chapter a precise name for this part of the problem.

The central question in term loans and the features of financing is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

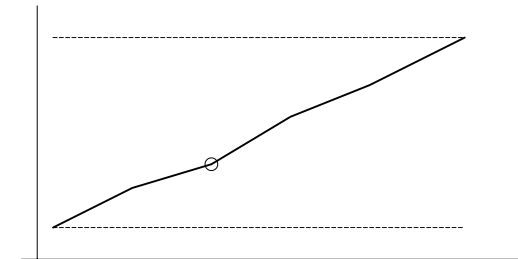


FIG. 15.1. Long-Term Financing and the Capital Market figure 1

15.1 TERM LOANS AND THE FEATURES OF FINANCING

The purpose here is to make term loans and the features of financing concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Term loan** means A loan repaid in regular instalments over a fixed number of years, usually from a bank. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Collateral** means An asset a lender may take if the borrower fails to repay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Covenant** means A condition in a loan agreement that the borrower promises to keep. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For discounting a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

5.2 MIXED STREAMS OF CASH FLOWS

The purpose here is to make mixed streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Discount rate** means The yearly percentage used in discounting; it states the return you could earn elsewhere. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash inflow** means Money coming into a business or to an investor, shown as a positive number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For mixed streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

The central question in discounting a single sum is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of mixed streams of cash flows before moving to application.

Discount rate. The yearly percentage used in discounting; it states the return you could earn elsewhere..

The term discount rate gives the chapter a precise name for this part of the problem.

Cash inflow. Money coming into a business or to an investor, shown as a positive number..

The term cash inflow gives the chapter a precise name for this part of the problem.

The central question in mixed streams of cash flows is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Working Capital and Short-Term Financing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Toyota's just-in-time system, developed from the late 1940s by Taiichi Ohno and others to cut stock held. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

- 14.T1 say what working capital is and why a profitable firm can still run short
- 14.T2 compute the cost of giving up a cash discount for new terms and recommend the cheaper source
- 14.T3 find the cash conversion cycle and the financing need for new data
- 14.T4 define float and its parts in plain words

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Present Value and Discounting is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ $(1 + \text{rate})$. Steps: Tk 100,000 $\times$ $(1 + 0.10)$ = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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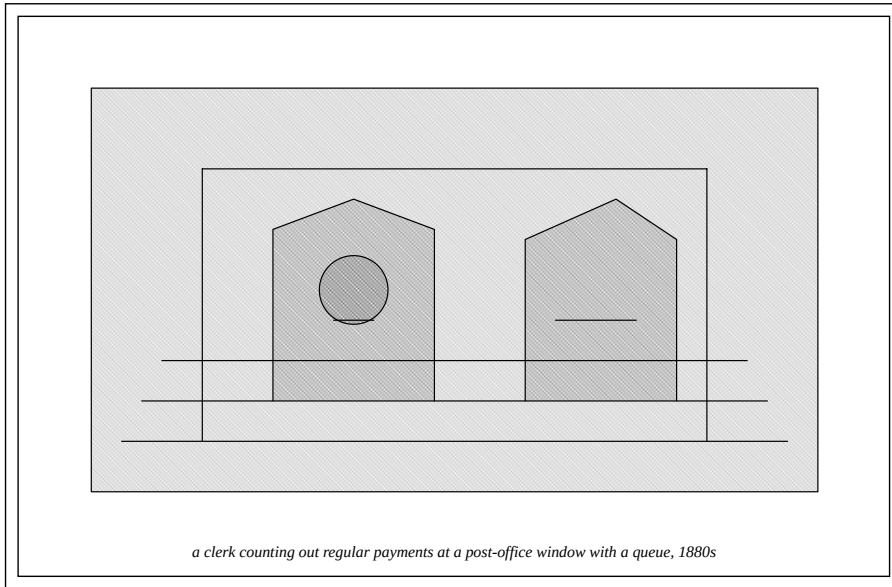
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CHAPTER VI

Annuities

A practical question about annuities



This chapter introduces annuities for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Long-term financing. Money raised for more than about five years, or with no fixed repayment date..

The term long-term financing gives the chapter a precise name for this part of the problem.

Trade credit. Credit from a supplier who lets a buyer pay some days after delivery..

The term trade credit gives the chapter a precise name for this part of the problem.

Cash discount. A price cut a supplier gives for paying early, such as 2% off if paid within 10 days..

The term cash discount gives the chapter a precise name for this part of the problem.

Inventory. The goods a business holds for sale, such as the books on a shop's shelves..

The term inventory gives the chapter a precise name for this part of the problem.

Accounts receivable. Money that customers owe a business for goods already delivered..

The term accounts receivable gives the chapter a precise name for this part of the problem.

The central question in sources of short-term finance is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of cycles, float and estimating needs before moving to application.

In this chapter, **Long-term financing** means Money raised for more than about five years, or with no fixed repayment date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Trade credit** means Credit from a supplier who lets a buyer pay some days after delivery. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash discount** means A price cut a supplier gives for paying early, such as 2% off if paid within 10 days. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Inventory** means The goods a business holds for sale, such as the books on a shop's shelves. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Accounts receivable** means Money that customers owe a business for goods already delivered. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For sources of short-term finance, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

14.3 CYCLES, FLOAT AND ESTIMATING NEEDS

The purpose here is to make cycles, float and estimating needs concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what an annuity is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.2 FUTURE VALUE OF AN ANNUITY

The purpose here is to make future value of an annuity concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Ordinary annuity** means An annuity whose payments fall at the end of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For future value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.3 PRESENT VALUE OF AN ANNUITY

The purpose here is to make present value of an annuity concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of future value of an annuity before moving to application.

Ordinary annuity. An annuity whose payments fall at the end of each period.

The term ordinary annuity gives the chapter a precise name for this part of the problem.

The central question in future value of an annuity is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of present value of an annuity before moving to application.

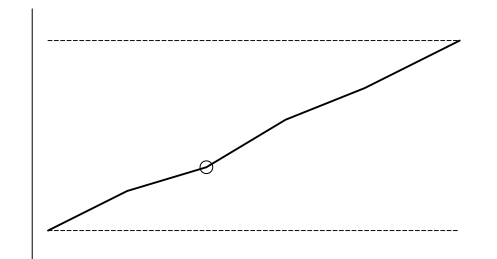


FIG. 14.1. Working Capital and Short-Term Financing figure 1

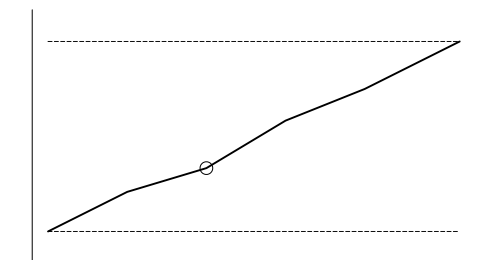


FIG. 14.2. Working Capital and Short-Term Financing figure 2

14.1 WORKING CAPITAL AND HOW TO FINANCE IT

The purpose here is to make working capital and how to finance it concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Working capital** means The money a business has tied up in everyday trading: cash, stock and money owed to it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of working capital and how to finance it before moving to application.

Working capital. The money a business has tied up in everyday trading: cash, stock and money owed to it.

The term working capital gives the chapter a precise name for this part of the problem.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to annuities. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

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- 13.T2 value a preferred stock and a growing common stock from new data
13.T3 explain why growth raises value but also needs a required return above growth

PAUSE AND CHECK 13.P1 After common stock, state the rule in your own words.

PAUSE AND CHECK 13.P2 After perpetuity, state the rule in your own words.

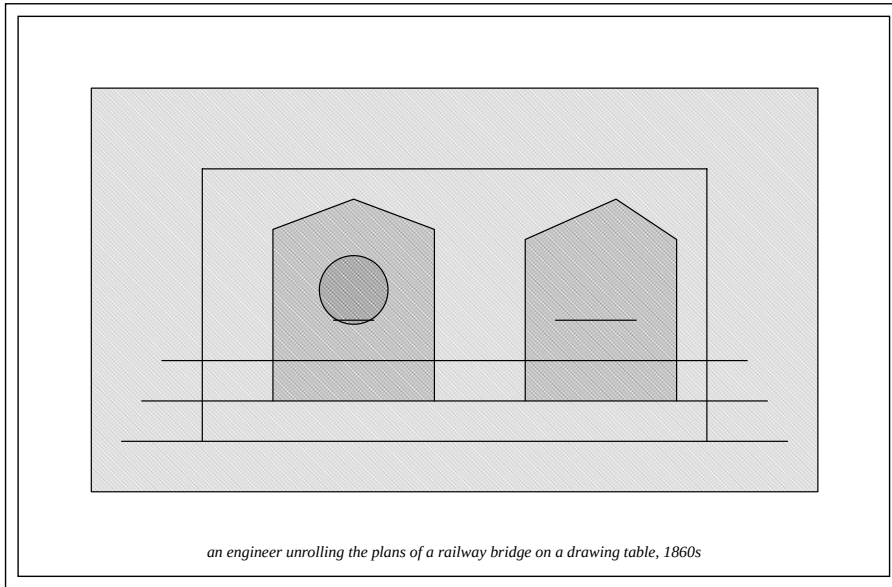
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AN ELEMENTARY INTRODUCTION TO MICROECONOMICS

A short restatement keeps the financial meaning attached to the calculation.

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CHAPTER VII

Capital Budgeting: Process and Cash Flows

A practical question about capital budgeting: process and cash flows



This chapter introduces capital budgeting: process and cash flows for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Constant growth model. The dividend valuation model for dividends that grow at the same rate every year..

The term constant growth model gives the chapter a precise name for this part of the problem.

The central question in valuing common stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

In this chapter, **Constant growth model** means The dividend valuation model for dividends that grow at the same rate every year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing common stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of shares. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

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The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For why firms budget for capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

7.2 THE STEPS OF THE PROCESS

The purpose here is to make the steps of the process concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Relevant cash flow** means A cash flow that changes because of the decision being studied. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sunk cost** means Money already spent that cannot be recovered whatever you decide now. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Initial investment** means The cash flows at the start of a project to buy and prepare the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Operating cash flow** means The cash a project brings in each year after paying its cash costs and tax. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the steps of the process before moving to application.

Relevant cash flow. A cash flow that changes because of the decision being studied.

The term relevant cash flow gives the chapter a precise name for this part of the problem.

Sunk cost. Money already spent that cannot be recovered whatever you decide now.

The term sunk cost gives the chapter a precise name for this part of the problem.

Initial investment. The cash flows at the start of a project to buy and prepare the asset.

The term initial investment gives the chapter a precise name for this part of the problem.

Operating cash flow. The cash a project brings in each year after paying its cash costs and tax.

The term operating cash flow gives the chapter a precise name for this part of the problem.

The central question in the steps of the process is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

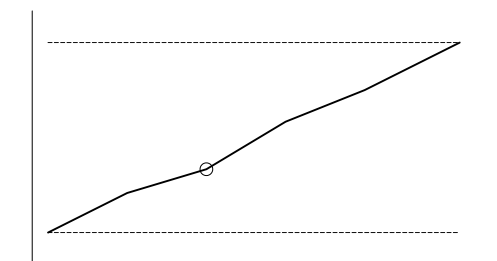


FIG. 13.1. Valuation of Shares figure 1

13.1 COMMON STOCK

The purpose here is to make common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Common stock** means Shares that give owners a vote and a claim on profit after all other claims are met. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Residual claim** means The right to whatever is left after lenders and preferred owners have been paid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For common stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

This section builds the idea of common stock before moving to application.

Common stock. Shares that give owners a vote and a claim on profit after all other claims are met.

The term common stock gives the chapter a precise name for this part of the problem.

Residual claim. The right to whatever is left after lenders and preferred owners have been paid.

The term residual claim gives the chapter a precise name for this part of the problem.

The central question in common stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

British consols, perpetual bonds first issued in the 1750s, and the UK government's redemption of its 3.5% War Loan in 1915. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognizable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

12.T1 explain what a bond is and name its main features with a new issue

12.T2 compute the value of three new bonds paying interest yearly and say whether each is at a premium, discount or par

12.T3 explain the link between coupon rate, required return and price

12.T4 define yield to maturity in plain words

PAUSE AND CHECK 12.P1 After bond features, state the rule in your own words.

PAUSE AND CHECK 12.P2 After pricing, state the rule in your own words.

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SUMMARY

The central purpose of Capital Budgeting: Process and Cash Flows is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

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Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

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IN HISTORY

The Concorde programme: UK and France agreed in 1962 to build it and continued after costs soared, a famous example of sunk-cost thinking; service ended in 2003. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

- 07.T1 say why capital budgeting matters to a business, with a new example
- 07.T2 list the steps of the process in order and what each step produces
- 07.T3 sort a list of costs into relevant and sunk

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

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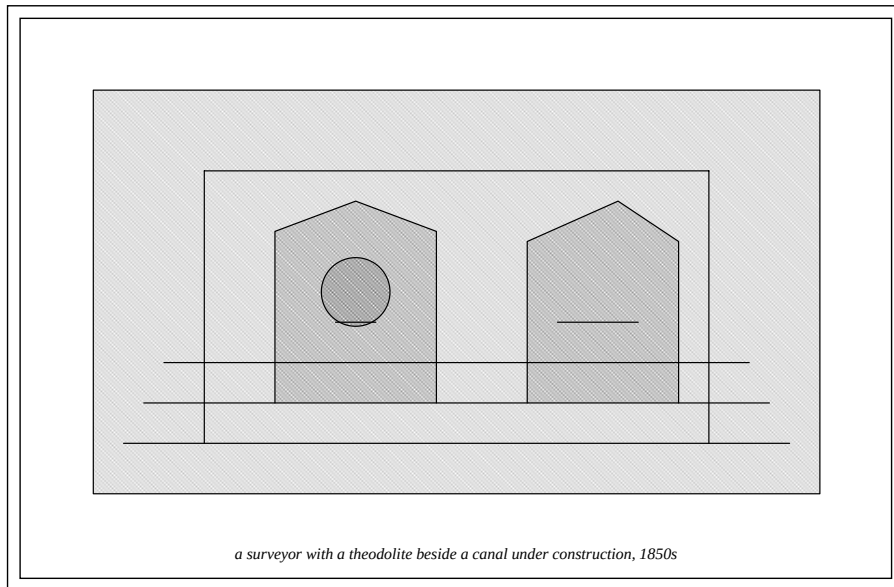
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CHAPTER VIII

Evaluating Projects: Payback and Net Present Value

A practical question about evaluating projects: payback and net present value



This chapter introduces evaluating projects: payback and net present value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

The central question in what a bond is is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of pricing a bond before moving to application.

Premium bond. A bond that sells for more than its face value..

The term premium bond gives the chapter a precise name for this part of the problem.

Discount bond. A bond that sells for less than its face value..

The term discount bond gives the chapter a precise name for this part of the problem.

Semiannual interest. Interest paid twice a year, each payment being half the yearly amount..

The term semiannual interest gives the chapter a precise name for this part of the problem.

The central question in pricing a bond is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what a bond is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.3 PRICING A BOND

The purpose here is to make pricing a bond concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Premium bond** means A bond that sells for more than its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discount bond** means A bond that sells for less than its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Semiannual interest** means Interest paid twice a year, each payment being half the yearly amount. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For pricing a bond, keep the decision in view. Ask what the business is trying to buy,

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

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A calculation can be correct and still answer the wrong question. For payback period, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

8.2 NET PRESENT VALUE

The purpose here is to make net present value concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Net present value** means The present value of all a project's future cash flows minus its initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Internal rate of return** means The discount rate at which a project's net present value is exactly zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For net present value, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

The central question in payback period is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of net present value before moving to application.

Net present value. The present value of all a project's future cash flows minus its initial investment..

The term net present value gives the chapter a precise name for this part of the problem.

Internal rate of return. The discount rate at which a project's net present value is exactly zero..

The term internal rate of return gives the chapter a precise name for this part of the problem.

The central question in net present value is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

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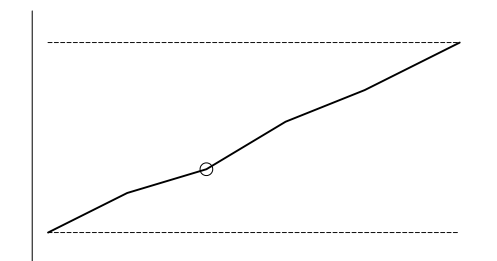


FIG. 12.1. Valuation of Bonds figure 1

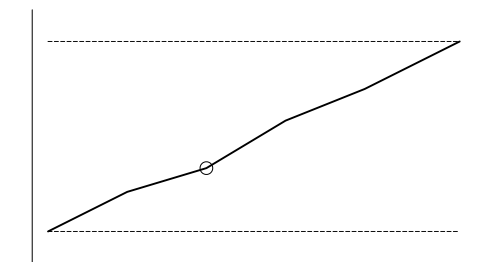


FIG. 12.2. Valuation of Bonds figure 2

12.1 THE IDEA OF VALUATION

The purpose here is to make the idea of valuation concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Valuation** means Finding what an asset is worth today by adding the present values of the cash it will bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of the idea of valuation before moving to application.

Valuation. Finding what an asset is worth today by adding the present values of the cash it will bring..

The term valuation gives the chapter a precise name for this part of the problem.

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* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to evaluating projects: payback and net present value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ $(1 + \text{rate})$. Steps: Tk 100,000 $\times$ $(1 + 0.10) =$ Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

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IN HISTORY

William Sharpe's 1964 paper on capital asset prices, and the 1990 Nobel Prize in economics shared by Markowitz, Miller and Sharpe. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

SOURCES

Chapter 9, 'Risk and Return', section 9.3 'Measuring risk' — standard deviation as the risk measure; Chapter 13, 'Valuation of Shares', section 13.3 'Valuing common stock' — required return used as the discount rate; Chapter 15, 'Long-Term Financing and the Capital Market', section 15.4 'The cost of capital' — equity cost..

REVIEW QUESTIONS

11.T1 explain risk premium and why investors demand it, with new figures

11.T2 separate systematic from unsystematic risk using a new list of risks

11.T3 compute required return for three new betas

11.T4 describe the three forms of market efficiency and what each implies

PAUSE AND CHECK 11.P1 After risk premium, state the rule in your own words.

PAUSE AND CHECK 11.P2 After beta, state the rule in your own words.

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SUMMARY

The central purpose of Evaluating Projects: Payback and Net Present Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

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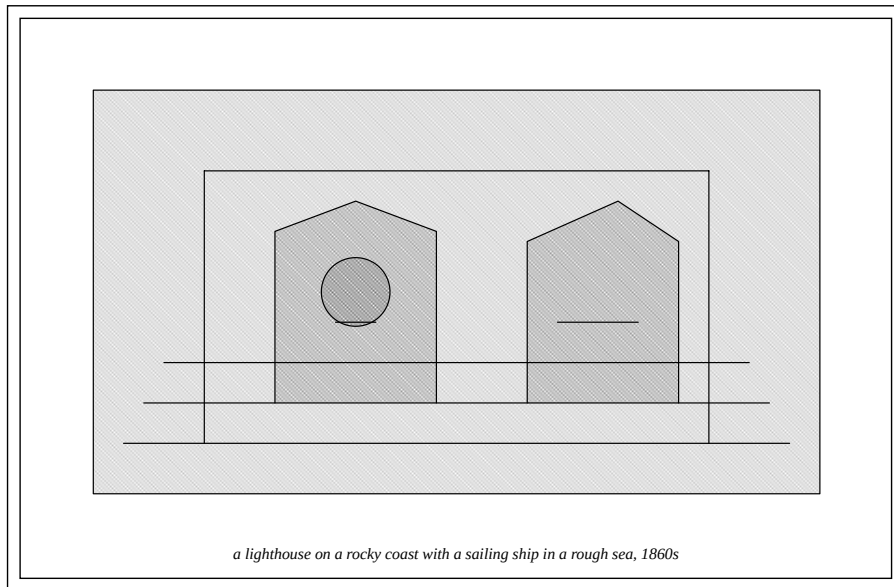
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A short restatement keeps the financial meaning attached to the calculation.



CHAPTER IX

Risk and Return

A practical question about risk and return



This chapter introduces risk and return for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the capital asset pricing model before moving to application.

Unsystematic risk. Risk that affects one firm or industry only; diversification can remove it..

The term unsystematic risk gives the chapter a precise name for this part of the problem.

Beta. A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market..

The term beta gives the chapter a precise name for this part of the problem.

Market return. The average return of all investments in a market during a period..

The term market return gives the chapter a precise name for this part of the problem.

The central question in the capital asset pricing model is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For diversification and beta, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.3 THE CAPITAL ASSET PRICING MODEL

The purpose here is to make the capital asset pricing model concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Unsystematic risk** means Risk that affects one firm or industry only; diversification can remove it. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Beta** means A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Market return** means The average return of all investments in a market during a period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the capital asset pricing model, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a

In this chapter, **Risk** means The chance that the actual result will differ from the result you expect. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Business risk** means Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For return and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

9.2 DESCRIBING UNCERTAINTY

The purpose here is to make describing uncertainty concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial risk** means Extra risk that comes from borrowing, because interest must be paid whatever the sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Probability distribution** means A list of possible results with the chance of each, the chances adding to one. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Risk. The chance that the actual result will differ from the result you expect..

The term risk gives the chapter a precise name for this part of the problem.

Business risk. Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing..

The term business risk gives the chapter a precise name for this part of the problem.

The central question in return and risk is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of describing uncertainty before moving to application.

Financial risk. Extra risk that comes from borrowing, because interest must be paid whatever the sales..

The term financial risk gives the chapter a precise name for this part of the problem.

Probability distribution. A list of possible results with the chance of each, the chances adding to one..

The term probability distribution gives the chapter a precise name for this part of the problem.

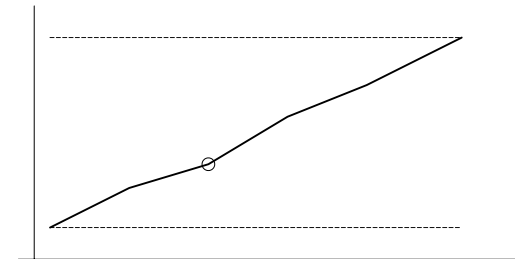


FIG. 11.1. Required Return, CAPM and Market Efficiency figure 1

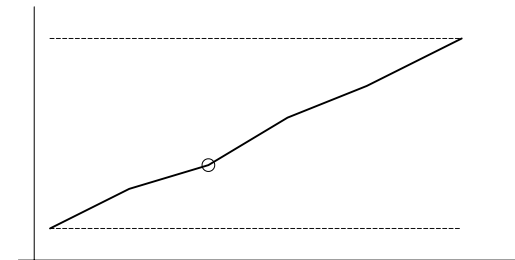


FIG. 11.2. Required Return, CAPM and Market Efficiency figure 2

11.1 RISK PREMIUM AND REQUIRED RETURN

The purpose here is to make risk premium and required return concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Required return** means The least return an investor will accept for taking a given risk. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Risk-free rate** means The return on an investment with no risk of loss, such as a short-term government security. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of risk premium and required return before moving to application.

Required return. The least return an investor will accept for taking a given risk..

The term required return gives the chapter a precise name for this part of the problem.

Risk-free rate. The return on an investment with no risk of loss, such as a short-term government security..

The term risk-free rate gives the chapter a precise name for this part of the problem.

The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For measuring risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to risk and return. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

IN HISTORY

Long-Term Capital Management, whose heavy borrowing led to a rescue organised by the Federal Reserve Bank of New York on 23 September 1998. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

10.T1 say what leverage means and tell its three kinds apart with new data

10.T2 find the break-even point and degrees of operating, financial and total leverage for a new firm

10.T3 explain in plain words why high leverage raises risk

PAUSE AND CHECK 10.P1 After break-even point, state the rule in your own words.

PAUSE AND CHECK 10.P2 After degree of operating leverage, state the rule in your own words.

PAUSE AND CHECK 09.P1 After probability distribution, state the rule in your own words.

PAUSE AND CHECK 09.P2 After standard deviation, state the rule in your own words.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of financial leverage before moving to application.

Operating leverage. The effect of fixed operating costs on how EBIT responds to sales.

The term operating leverage gives the chapter a precise name for this part of the problem.

Financial leverage. The effect of fixed financing payments on how owners' earnings respond to EBIT.

The term financial leverage gives the chapter a precise name for this part of the problem.

Total leverage. The combined effect of operating and financial leverage on how owners' earnings respond to sales.

The term total leverage gives the chapter a precise name for this part of the problem.

The central question in financial leverage is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For operating leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

10.3 FINANCIAL LEVERAGE

The purpose here is to make financial leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating leverage** means The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the percentage change in EBIT divided by the percentage change in sales. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

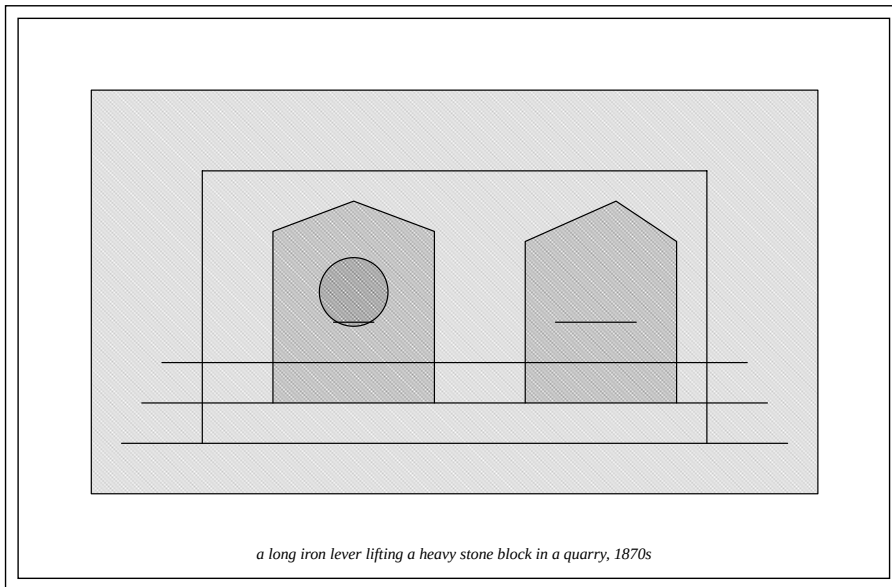
In this chapter, **Financial leverage** means The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Total leverage** means The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For financial leverage, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a



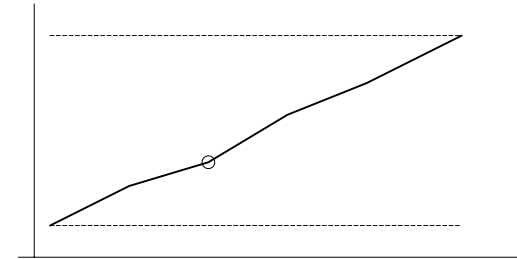
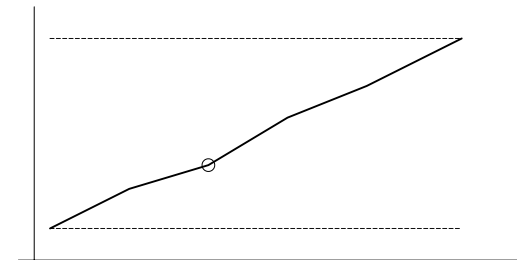
CHAPTER X

Leverage

A practical question about leverage



This chapter introduces leverage for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

FIG. 10.1. *Leverage figure 1*FIG. 10.2. *Leverage figure 2*

10.1 COSTS AND THE BREAK-EVEN POINT

The purpose here is to make costs and the break-even point concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Leverage** means The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Fixed cost** means A cost that stays the same however many units are sold, such as shop rent. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of costs and the break-even point before moving to application.

Leverage. The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings..

The term leverage gives the chapter a precise name for this part of the problem.

Fixed cost. A cost that stays the same however many units are sold, such as shop rent..

The term fixed cost gives the chapter a precise name for this part of the problem.